

Mindala – LUMA 2026 Small Works Application

Biofeedback-Responsive Immersive Dome Installation

Robert F. Thomas (Metta)

June 29, 2026

GitHub Repository (Full Documentation + Live Functional Demo):

<https://github.com/IAmM3ta/mindala>

Refined Artist Statement

Robert F. Thomas (Metta) is a projection artist and applied physicist who conceptualized **Mindala** in 2015 while working in the aerospace engineering and materials science laboratory at TU Delft. He built a working small-scale prototype using one Emotiv EPOC EEG headset and a custom cymatics setup. After rising to become a leading projection mapper for major international festivals, Hurricane Helene destroyed his studio and approximately \$100,000 worth of specialized AV equipment, leaving him marooned in a disaster zone and unable to physically implement projects.

Mindala revives that long-held vision: a biofeedback-responsive dome that translates real-time EEG and physiological data into living 360° mandala visualizations. Participants become the engine of the work as their brainwaves and heart rhythms shape symmetrical geometry around them. This transforms projection mapping into active neural entrainment and self-revelation — not spectacle, but profound emotional and physiological connection.

Character count: 498

LUMA Small Works Application Responses

2.1 Artist Statement

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2.2 Artwork/Project Title

Mindala

2.3 Year Completed

2026 (proposed for LUMA; originally conceptualized 2015 with working prototype)

2.4 Artwork/Project Description

Mindala is a biofeedback-responsive immersive dome installation that creates a closed cybernetic loop between participants and their environment. Guests recline in sensor-equipped chairs inside a seamless negative-pressure projection dome. Real-time EEG (Emotiv EPOC) and biometric data (HRV, respiration, EDA) drive the proprietary VUSIC engine, generating 360° mandala visualizations and isochronic spatial audio.

The experience unfolds in five phases: Arrival, Grounding, Attention, Coherence (where visuals crystallize in perfect symmetry with the participant's physiological state), and Release. Participants receive a personalized "mandala signature" digital artwork.

The system supports interior + optional exterior mapping, turning the dome into both an intimate therapeutic space and a luminous public beacon. Mindala directly answers LUMA Small Works' call for experimental, immersive, tech-based installations that forge deep emotional connections.

2.5 Artwork/Project Tech

Existing assets I bring: M1 Max MacBook Pro (64GB RAM, 32-core), Epson L610U laser projector, Sony VPL-PHZ51 laser, Optoma UHD60 4K, Epson Home Cinema 5030UB, 2× Panasonic VZ580 (bulbs needed), and 3× Zoom VR microphones.

Core additional requirements: 1–3 Emotiv EPOC X EEG headsets, dedicated GPU workstation (Mac Studio or equivalent), seamless negative-pressure fulldome, custom biofeedback supine chairs (PPG/HRV, respiration, EDA sensors), and immersive spatial audio system.

The VUSIC engine handles real-time signal processing, feature extraction, OSC/MIDI sonification, cymatics chamber integration, LSTM-stabilized generative shaders, and fulldome spherical mapping with edge-blending. Fully compatible with Resolume, TouchDesigner, MadMapper, and Synesthesia for layer mixing. System includes deterministic fallback masters for reliability.

Full technical details and working pipeline demo: <https://github.com/IAmM3ta/mindala>

2.6 Which of the following best describes... (Select all that apply)

Generative Art, Projection Mapping, Art and Technology, Sound Installation

2.7 Portfolio Website

<https://github.com/IAmM3ta/mindala>

2.8 Vimeo/Youtube/Webpage link... (proof-of-concept)

<https://github.com/IAmM3ta/mindala>

2.9 Vimeo/Youtube/Webpage Password (if needed)

(none)

2.10 Are you able to attend LUMA Projection Arts Festival 2026 in person?

Yes

2.11 How did you learn about LUMA Projection Arts Festival?

Through the official LUMA Small Works open call announcement and immersive arts community networks.

Detailed Technical Requirements, Budget & Timeline

3.1 Equipment I Already Own

- M1 Max MacBook Pro (64GB RAM, 32 cores, 4TB SSD)
- Epson L610U laser projector (primary)
- Sony VPL-PHZ51 laser projector
- Optoma UHD60 4K + Epson Home Cinema 5030UB
- 2× Panasonic VZ580 WUXGA (bulbs needed)
- 3× Zoom VR Microphones

3.2 Additional Equipment Required

Item	Quantity	Estimated Cost
Emotiv EPOC X EEG Headset	1–3	\$999 – \$3,000
Dedicated GPU Workstation	1	\$2,200 – \$3,800
Seamless negative-pressure full-dome (4–6m)	1	\$4,500 – \$9,500
Biofeedback sensors + custom chairs	3-chair kit	\$1,800 – \$3,800
Immersive spatial audio system	1	\$2,200 – \$5,000
Subtotal – Production		\$11,700 – \$25,100

3.3 Artist Travel & Personal Expenses (Realistic 18-Day Build + Festival Period)

Expense	Estimated Cost
Round-trip travel (Asheville, NC – Bing-hamton, NY)	\$320 – \$480
Accommodation (18 nights)	\$2,160 – \$2,880
Meals (18 days × \$70/day)	\$1,260
Local transportation + incidentals	\$350 – \$500
Subtotal – Personal Expenses (18 days)	\$4,090 – \$5,120

Note: LUMA provides travel and lodging stipends for artists traveling more than 75 miles. These costs may be partially offset.

3.4 Total All-In Budget

Tier	Production	Personal (18 days)	Total
Minimal Viable	\$8,500–12,000	\$4,090–5,120	\$12,600–17,120
Recommended	\$15,000–22,000	\$4,090–5,120	\$19,100–27,120
Full Flagship	\$25,000–35,000	\$4,090–5,120	\$29,100–40,120

3.5 Revised Timeline

- **Now – July 1:** Submit application + sponsorship outreach
- **Mid-July – Mid-August:** Notification + remote preparation
- **Late August 2026** (arrive Aug 28): Begin on-site build
- **Late August – Sept 9:** Dome setup, sensor integration, full system testing (10–14 days)
- **Sept 10:** Final tech rehearsal
- **Sept 11–12:** Festival
- **Sept 13–14:** Teardown + departure

Resources

GitHub Repository: <https://github.com/IAMM3ta/mindala> (Contains full documentation + working Python demo of the EEG → Sonification → Cymatic → Video Mapping pipeline with mixing examples for Resolume/TouchDesigner/etc.)